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DO NOT TRAIN AWAY UNCERTAINTY: EARLY UNCERTAINTY ANCHORED CALIBRATION

Yutong Xie[1], Jiawei Tang[2], Zhenglin Hua[2], Yuxiang Ma[1], Si Qin[2], Yaxin Hou[2], Hui Liu[4], Junhui Hou[5], Yuheng Jia[2,3,4]*
[1]School of Software Engineering, Southeast University, Nanjing, China
[2]School of Computer Science and Engineering, Southeast University, Nanjing, China
[3]Key Laboratory of New Generation Artificial Intelligence Technology and Its Interdisciplinary Applications (Southeast University), Ministry of Education, China
[4]School of Computing Information Sciences, Saint Francis University, Hong Kong, China
[5]Department of Computer Science, City University of Hong Kong, Hong Kong, China
{xyt,jw_tang,220252358,220256453,siqin,yaxin,yhjia}@seu.edu.cn
h2liu@sfu.edu.hk, jh.hou@cityu.edu.hk

ABSTRACT

Deep neural networks, including large language models, have achieved remarkable performance across various tasks. However, they are prone to overconfidence during training or fine-tuning. In this work, we observe a consistent phenomenon across different models that the early model is better calibrated, while later training or fine-tuning yields marginal accuracy gains but substantially increases calibration errors. Our analysis suggests that the early model retains uncertainty awareness in both its predictions and features, which is gradually lost with continued training. To avoid training away this uncertainty awareness, we propose **EUA-Cal**, a novel method that exploits the **E**arly model as an **U**ncertainty **A**nchor for **Cal**ibration. EUA-Cal introduces early prediction regularization to preserve early predictive uncertainty and prototype structure regularization to exploit uncertainty reflected in the early feature space, jointly mitigating overconfidence. Extensive experiments on image classification and multiple-choice question answering across eight diverse models demonstrate that EUA-Cal outperforms state-of-the-art calibration methods.

1 INTRODUCTION

In recent years, deep neural networks (DNNs), including large language models (LLMs), have achieved significant progress in image classification (Zagoruyko & Komodakis, 2016), object segmentation (Kirillov et al., 2023), and natural language processing (Vaswani et al., 2017). Despite their remarkable performance, these models are prone to overconfidence during training or fine-tuning, with the predicted confidence exceeding the actual accuracy (Guo et al., 2017; Jiang et al., 2021). This overconfidence reflects an underestimate of the uncertainty in their predictions and can reduce their reliability, particularly in high-risk applications such as autonomous driving (Zhang et al., 2025) and medical diagnosis (Huang et al., 2025b).

To calibrate model confidence, various calibration techniques have been proposed, including post-hoc calibration (Guo et al., 2017; Pan et al., 2020), loss function and regularization designs (Szegedy et al., 2016; Mukhoti et al., 2020; Liu et al., 2022; Ni et al., 2025), mixup-based methods (Thulasidasan et al., 2019; Bouniot et al., 2025), uncertainty-aware methods (Sensoy et al., 2018; Li et al., 2025), etc. Expected Calibration Error (ECE) (Guo et al., 2017) is a commonly used metric for evaluating the calibration performance of these methods, which measures the discrepancy between predicted confidence and empirical accuracy across confidence bins. A lower value of ECE indicates better calibration performance, i.e., the model is aware of how uncertain its predictions are.

*Corresponding author.